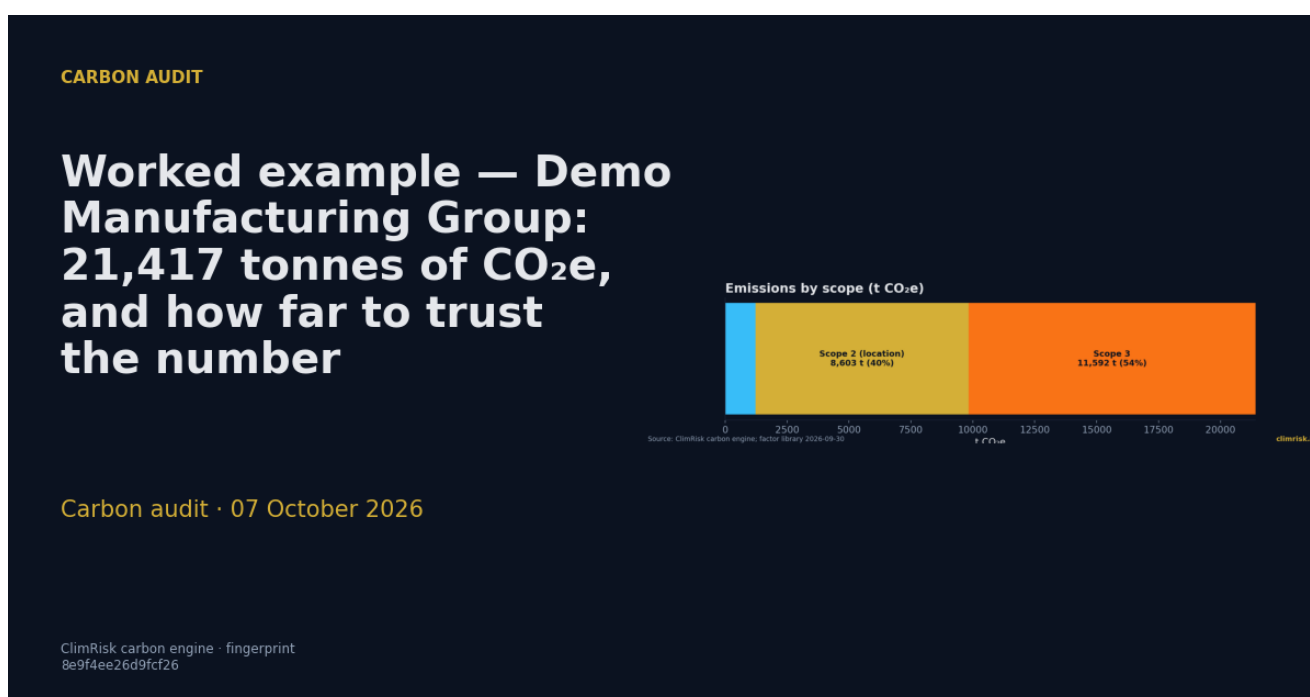


Worked example — Demo Manufacturing Group: 21,417 tonnes of CO₂e, and how far to trust the number

Reporting year 2025, equity share consolidation, GWP basis AR6. Scope 1 1,222 t, scope 2 8,603 t (location) or 9,486 t (market), scope 3 11,592 t. 13 of 21 categories have no data.

ClimRisk carbon engine · 2026-10-07



2025. An emissions number is only as good as the lines behind it. For Worked example — Demo Manufacturing Group the engine audits 19 lines into 21,417 t CO₂e; Purchased electricity is 40% of it, and 13 of 21 categories carry no data at all [1].

Worked example — Demo Manufacturing Group in six numbers

A carbon inventory, audited line by line

1,222 t

scope 1

direct emissions

8,603 t

scope 2, location-based

grid-average factors

9,486 t

scope 2, market-based

contractual / residual-mix factors

11,592 t

scope 3 reported

8 of 21 categories

2.89

data-quality score

1 best, 5 worst

47%

primary-data share

the rest uses spend or default factors

Source: ClimRisk carbon engine; GHG Protocol; factor library 2026-09-30

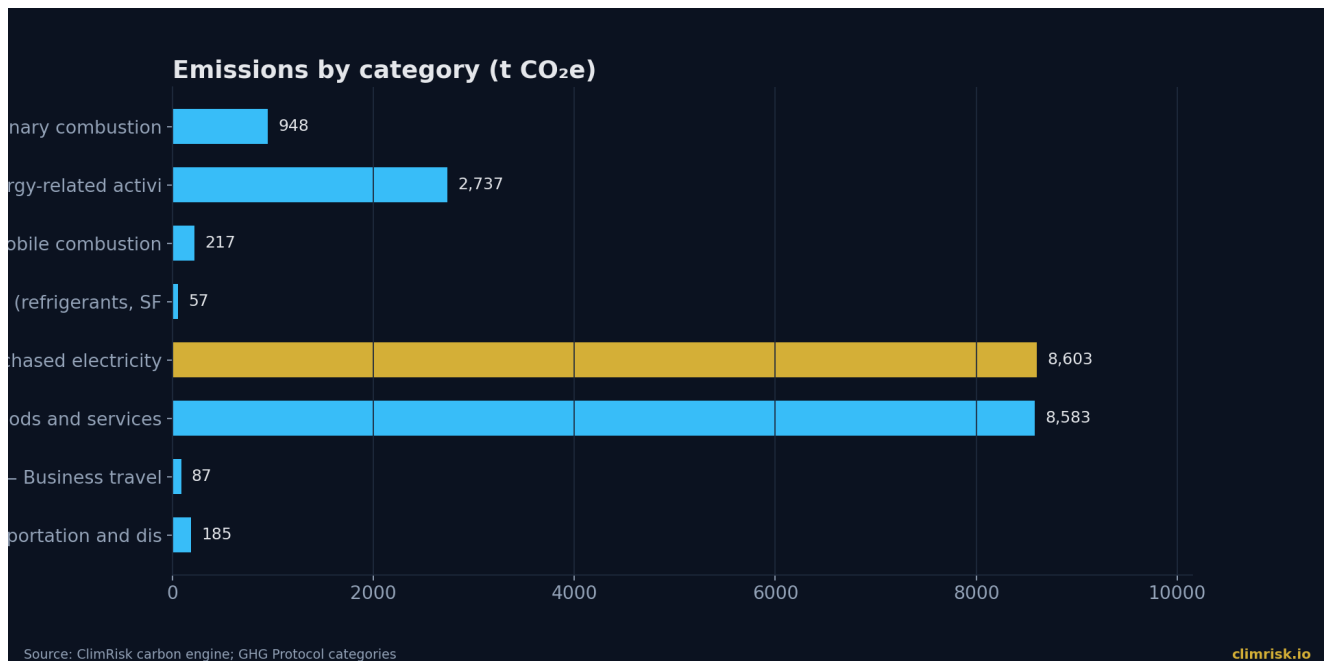
climrisk.io

Each number is a sum of audited lines.

13 categories with no data

Purchased electricity carries 40% of the reported total. A category with no data must be explained, not dropped; the auditor will ask first.

The loud number



Every bar is the sum of audited lines; each line keeps its input, factor and evidence reference.

Intensity: 23.4 t per US\$ m revenue and 5.46 t per employee (Scope 1 + Scope 2 (location-based)).

The quiet part: what is missing, and what an auditor will ask

13 of 21 categories have no data. Process emissions (none); Purchased heat, steam and cooling (none); Cat 2 — Capital goods (excluded); Cat 5 — Waste generated in operations (excluded); Cat 7 — Employee commuting (excluded); Cat 8 — Upstream leased assets (not_relevant).

Level	Check		Message			
warning	method		record r6: eGRID subregion inferred from state TX (87% of in-state generation in ERECT); give the subregion for exactness			
warning	method		record r7: market-based: no residual mix for this grid, location-based factor used			
warning	method		record r7: No official national factor in the library — supply the national/utility factor if available			
Line	Entity	Scope	t CO ₂ e	Factor	Evidence	DQ
Steel purchases	Demo Inc (Texas)	3	8,583.5	useio13:331110	AP ledger	4
Plant electricity	Demo India Pvt	2	5,485.5	grid	DISCOM bills	2
Plant electricity	Demo Inc (Texas)	2	2,272.6	grid	Retailer invoices	2
Plant electricity	Demo India Pvt	3	995.4	td_losses	DISCOM bills	3
Boilers	Demo UK Ltd	1	948.1	desnz26:fuels-gaseous-fuels-na	Gas invoices 2025	1
Plant electricity	Demo India Pvt	3	549.0	ember_upstream	DISCOM bills	3
Plant electricity	Demo Inc (Texas)	3	545.4	ember_upstream	Retailer invoices	3
Plant electricity	Demo GmbH (60%)	2	439.1	grid	Stadtwerke invoices	2

Largest lines. A line with no evidence reference will be queried first [1].

What this means, by reader

For the sustainability lead. The completeness table is the work list before assurance.

For the assurer. Fingerprint 8e9f4ee26d9fcf26 reproduces the calculation from the same inputs and factor library.

Method and limitations

Scope 2 is reported both ways; market-based uses residual-mix factors where supplied. Primary data cover 47% of the total; the rest uses spend or default factors. GHG Protocol Corporate Standard (rev.) GHG Protocol Scope 2 Guidance (2015) GHG Protocol Corporate Value Chain (Scope 3) Standard

How ClimRisk measured this

- Inputs → activity data per entity, scope and category, each with an evidence reference.
- Factors → the ClimRisk emission-factor library, versioned, with named sources.
- Checks → completeness, consistency and data-quality rules.
- Output → totals by scope and category and a reproducible fingerprint.

Sources

1. ClimRisk emission-factor library 2026-09-30 — UK DESNZ 2026, US EPA eGRID 2023, Green-e residual mix, AIB, Ember

Produced by the ClimRisk engine from the sources listed. Model outputs with the limitations stated; not forecasts or investment advice.